

Alex Junior Gomez Saltachin

PhD Student in Mathematics, PUC-Rio, Brazil

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Research Interests

Arithmetic Geometry, Homological Mirror Symmetry, Periods of Landau–Ginzburg Models, Topological Quantum Field Theory, Modular Forms, and Geometric Representation Theory.

Education

- **PhD in Mathematics**, PUC-Rio, Brazil, 2023–present.
Advisor: Sergey Galkin.
- **MSc in Mathematics**, Pontificia Universidad Católica del Perú, 2019.
Thesis: *Modular Forms*.
- **BSc in Mathematics**, Pontificia Universidad Católica del Perú, 2015.
Thesis: *Elliptic Curves: Mordell–Weil, Roth and Siegel Theorems*.

Selected Schools and Programs

- ICTP School and Workshop on Explicit Arithmetic Geometry, ICTP, 2025.
- Conference on Recent Developments in Topological Quantum Field Theory, BIMSA, 2024.
- The Mordell Conjecture 100 Years Later, MIT, 2024.
- ICTP String-Math, ICTP, 2024.
- CIMPA ELGA – Latin American School of Algebraic Geometry, 2024.
- Arizona Winter School, 2022 and 2023.
- IHES Summer School on the Langlands Program, 2022.
- ICTP–CIMPA Schools in Arithmetic Geometry, 2015 and 2018.

Selected Talks

- *Periods in Homological Mirror Symmetry and Kloosterman Motives*, ICTP, 2025.
- *Mordell–Weil Theorem*, PUC-Rio Friday Seminar, 2024.
- *Karoubian Completion of Triangulated Categories*, PUC-Rio Friday Seminar, 2024.

Teaching Experience

Graduate Instructor and Teaching Assistant at PUC-Rio and PUCP, 2013–2026.

Courses taught or assisted include Linear Algebra, Algebra, Real Analysis, Complex Analysis, Calculus, and Mathematics for Economists.

Academic Service

- Co-founder and organizer of the *38 Movimientos Seminar*.
- Co-organizer of the *Meetings on Arithmetic and Algebraic Number Theory Seminar*.

Languages

Spanish native; English advanced; Portuguese advanced; basic Chinese and French.

References

- **Sergey Galkin** — PhD Advisor, PUC-Rio.
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- **Misha Verbitsky** — IMPA.
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- **Sergei Burkin** — PUC-Rio.
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- **Alfredo Poirier** — MSc Advisor, PUCP.
apoirie@pucp.edu.pe
- **Rudy Rosas** — PUCP.
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- **Richard Gonzales** — PUCP.
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- **Jaime Cuadros** — PUCP.
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- **Percy Fernandez** — PUCP.
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